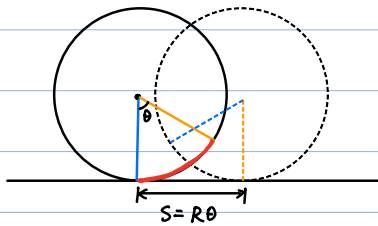


Angular Momentum

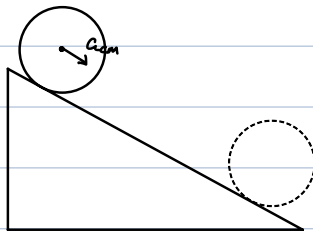
Rotation: fixed axis $\rightarrow$ moving axis

Pure rotation



$$v_{cm} = \frac{ds}{dt} = R \frac{d\theta}{dt} = R\omega$$

$$a_{cm} = \frac{dv_{cm}}{dt} = R\alpha$$



Q: $\vec{a}_{cm} = ?$

see later

Note: When point of contact remains static,
friction does No work in pure rotation

$\Rightarrow$ Mechanical energy conservation.

Total kinetic energy:

$$K = K_{cm} + K' \quad (\text{König theorem})$$

$$= \frac{1}{2} M v_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$$

$$= \frac{1}{2} \left(M + \frac{I_{cm}}{R^2} \right) v_{cm}^2 \quad (v_{cm} = \omega R \text{ for pure rotation})$$

We learnt the Impulse - Momentum Theorem :

$$\vec{I} = \vec{p}_f - \vec{p}_i$$

Similar thing for rotation?

Torque:

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\Rightarrow \vec{\tau} = \vec{r} \times \frac{d\vec{p}}{dt}$$

$$\Rightarrow \vec{\tau} = \frac{d}{dt} (\vec{r} \times \vec{p}) - \underbrace{\frac{d\vec{r}}{dt} \times \vec{p}}_{\vec{v} \times (m\vec{v}) = 0}$$

Angular Momentum:

$$\vec{L} = \vec{r} \times \vec{p}$$

$\Rightarrow$

$$\vec{\tau} dt = d\vec{L}$$

Notes:

1. Both $\vec{\tau}$ & $\vec{L}$ depend on choice of origin.
2. The origin of $\vec{\tau}$ & $\vec{L}$ should be the same.
3. Any origin is good.

A system of particles:

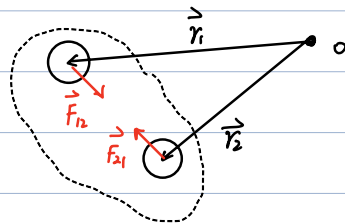
$$\begin{cases} \vec{L} = \sum_i \vec{L}_i \\ \vec{\tau}_{\text{ext}} \equiv \sum_i \vec{\tau}_{i,\text{ext}} \quad (\text{external}) \end{cases}$$

$$\Rightarrow \vec{\tau}_{\text{ext}} dt = d\vec{L}$$

Note:

Internal torques cancel

e.g.



$$\begin{aligned} \vec{\tau}_{\text{int}} &= \vec{r}_1 \times \vec{F}_{12} + \vec{r}_2 \times \vec{F}_{21} \\ &= (\vec{r}_1 - \vec{r}_2) \times \vec{F}_{12} \\ &= 0 \end{aligned}$$

$$(\vec{r}_1 - \vec{r}_2 \parallel \vec{F}_{12})$$

(Similar to Newton's 2nd law, only external forces have to be considered)

Rigid body rotation around a fixed axis:

$$\vec{L} = \sum_i \vec{r}_i \times m_i \vec{v}_i$$

(2)

$$= \sum_i \vec{r}_i \times m_i (\vec{\omega} \times \vec{r}_i)$$

$$= \sum_i m_i [r_i^2 \vec{\omega} - (\vec{r}_i \cdot \vec{\omega}) \vec{r}_i]$$

$$\vec{\omega} = \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix} \quad \vec{r}_i = \begin{pmatrix} x_i \\ y_i \\ z_i \end{pmatrix}$$

$$\Rightarrow \vec{L} = \sum_i m_i \begin{pmatrix} (y_i^2 + z_i^2) & -x_i y_i & -x_i z_i \\ -y_i x_i & (z_i^2 + x_i^2) & -y_i z_i \\ -z_i x_i & -z_i y_i & (x_i^2 + y_i^2) \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix}$$

I : Moment of inertia tensor

i.e. $\boxed{\vec{L} = I \vec{\omega}}$

where

$$\boxed{I = \sum_i m_i (r_i^2 \mathbf{1} - \vec{r}_i \otimes \vec{r}_i)}$$

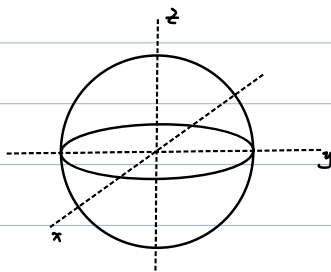
Notes:

1. $(\vec{A} \otimes \vec{B})_{mn} \equiv A_m B_n$ dyad product, or direct product

2. further check:

$$\begin{aligned} & \sum_i m_i [r_i^2 \vec{\omega} - (\vec{r}_i \cdot \vec{\omega}) \vec{r}_i] \\ &= \sum_i m_i [r_i^2 \omega_a - \sum_b r_{i,b} \omega_b r_{i,a}] \\ &= \sum_b \underbrace{\sum_i m_i (r_i^2 \delta_{ab} - r_{i,a} r_{i,b})}_{I_{ab}} \omega_b \end{aligned}$$

Example: uniform ball



$$I = \begin{pmatrix} I_0 & & \\ & I_0 & \\ & & I_0 \end{pmatrix}$$

$$I_0 = \frac{2}{5} MR^2$$

check: $I_{xy} = \int dx dy dz (-xy) \rho$

odd in x , odd in y

$$\Rightarrow I_{xy} = 0$$

If z -axis is fixed $\Rightarrow L_z = I_0 \omega_z$

(3)

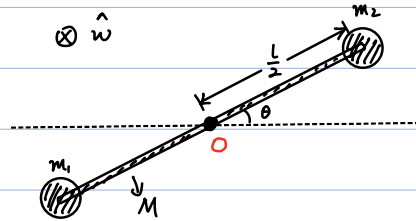
Example.

Q: $\vec{L}$? $\vec{\alpha}$?

$$I = \frac{1}{12} M L^2 + (m_1 + m_2) \frac{L^2}{4}$$

$$\Rightarrow L = I \omega$$

$$= \frac{L^2}{4} \left(\frac{1}{3} M + m_1 + m_2 \right) \omega$$



Note: origin at O.

$$\tau_{\text{ext}} = m_2 g \frac{L}{2} \cos \theta - m_1 g \frac{L}{2} \cos \theta$$

$$= I \alpha$$

$$\Rightarrow \alpha = \frac{(m_2 - m_1) g \frac{L}{2} \cos \theta}{\frac{L^2}{4} \left(\frac{1}{3} M + m_1 + m_2 \right)}$$

$$= \frac{2(m_2 - m_1) g \cos \theta}{L \left(\frac{M}{3} + m_1 + m_2 \right)}$$

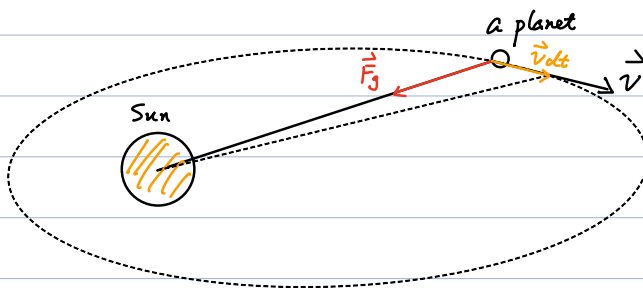
• Conservation of angular momentum:

$$\vec{\tau}_{\text{ext}} = 0$$

$$\Rightarrow \frac{d\vec{L}}{dt} = 0$$

$$\Rightarrow \vec{L} = \text{constant}$$

Example: Kepler's second law



$$\tau = \vec{r} \times \vec{F}_g = 0$$

$$\Rightarrow \vec{L} = \vec{r} \times \vec{p} = \text{constant}$$

$$\vec{r} \times \vec{p} = m \vec{r} \times \vec{v}$$

Area swept by $\vec{r}$ in unit time:

$$dA = \frac{1}{2} |\vec{r} \times \vec{v} dt|$$

$$\Rightarrow \frac{dA}{dt} = \frac{1}{2} |\vec{r} \times \vec{v}|$$

$$= \frac{|\vec{L}|}{2m} = \text{constant}$$

④

Center of mass:

$$\vec{r} = \vec{r}_{cm} + \vec{r}'$$

$\vec{r}'$: position relative to CM

$$\Rightarrow \vec{\tau}_{ext} = \sum_i \vec{r}_i \times \vec{F}_{i,ext}$$

$$= \sum_i \vec{r}_{cm} \times \vec{F}_{i,ext} + \sum_i \vec{r}'_i \times \vec{F}_{i,ext}$$

$$\Rightarrow \boxed{\vec{\tau}_{ext} = \vec{r}_{cm} \times \left(\sum_i \vec{F}_{i,ext} \right) + \sum_i \vec{r}'_i \times \vec{F}_{i,ext}}$$

all external forces on CM

torques relative to CM

Note: $\sum_i \vec{r}'_i \times m_i \vec{g} = \left(\sum_i m_i \vec{r}'_i \right) \times \vec{g} = 0$

$$\Rightarrow \vec{L} = \sum_i \vec{r}_i \times m_i \vec{v}_i$$

$$= \sum_i (\vec{r}_{cm} + \vec{r}'_i) \times m_i (\vec{v}_{cm} + \vec{v}'_i)$$

$$= \sum_i m_i \vec{r}_{cm} \times \vec{v}_{cm} + \sum_i m_i \vec{r}'_i \times \vec{v}'_i + \cancel{\sum_i m_i \vec{r}_{cm} \times \vec{v}'_i} + \cancel{\sum_i m_i \vec{r}'_i \times \vec{v}_{cm}}$$

Note: $\sum_i m_i \vec{r}'_i = \sum_i m_i \vec{r}_i - \sum_i m_i \vec{r}_{cm}$

$$= \left(\sum_i m_i \right) \vec{r}_{cm} - \sum_i m_i \vec{r}_{cm} = 0$$

$$\Rightarrow \boxed{\vec{L} = M \vec{r}_{cm} \times \vec{v}_{cm} + \sum_i m_i \vec{r}'_i \times \vec{v}'_i}$$

relative to CM

Combine the formulas of torque & angular momentum:

$$\sum_i \vec{r}'_i \times \vec{F}_{i,ext} = \frac{d}{dt} \left(\sum_i m_i \vec{r}'_i \times \vec{v}'_i \right)$$

$$\vec{r}_{cm} \times \left(\sum_i \vec{F}_{i,ext} \right) = \frac{d}{dt} \left(M \vec{r}_{cm} \times \vec{v}_{cm} \right)$$

Reminder: the impulse - momentum theorem in CM:

$$\boxed{\sum_i \vec{F}_{i,ext} = \frac{d}{dt} (M \vec{v}_{cm})}$$

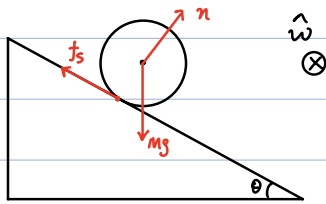
check: $\sum_i \vec{F}_{i,ext} = \frac{d}{dt} \sum_i m_i (\vec{v}_{cm} + \vec{v}'_i)$

$$= \frac{d}{dt} M \vec{v}_{cm} + \cancel{\frac{d^2}{dt^2} \sum_i m_i \vec{r}'_i}$$

• Generally, for a closed, isolated system,

$$\begin{cases} K_i + U_i = K_f + U_f \\ \vec{P}_i = \vec{P}_f \\ \vec{L}_i = \vec{L}_f \end{cases}$$

Back to the rolling ball:



Method-1 :

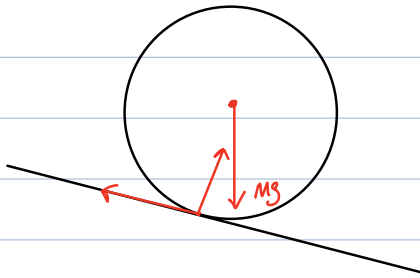
$$\begin{cases} Mg \sin \theta - f_s = Ma_{cm} & (\text{momentum}) \\ f_s R = I_{cm} \alpha & (\text{relative angular momentum}) \\ a_{cm} = \alpha R & (\text{pure rotation}) \end{cases}$$

$$\Rightarrow \alpha = \frac{Mg \sin \theta}{MR + I_{cm}/R}$$

$$\Rightarrow a_{cm} = \frac{g \sin \theta}{1 + I_{cm}/(MR^2)}$$

Method-2 :

$\vec{n}$ & $\vec{f}_s$ has no torque at the bottom



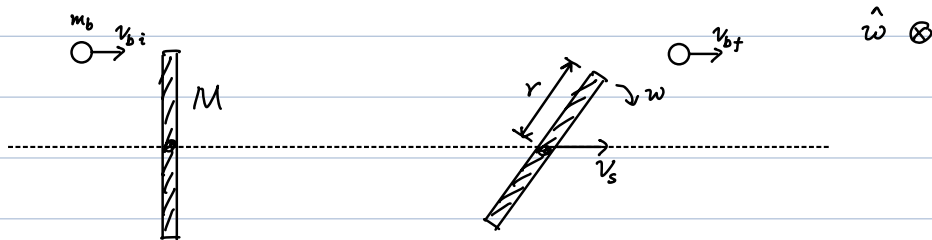
$$\tau_{ext} = Mg R \sin \theta$$

$$I = I_{cm} + MR^2 \quad (\text{parallel axis})$$

$$\Rightarrow Mg R \sin \theta = (I_{cm} + MR^2) \frac{a_{cm}}{R}$$

$$\Rightarrow a_{cm} = \frac{g \sin \theta}{1 + I_{cm}/(MR^2)}$$

Example : elastic collision between a ball and a stick



$$m_b v_{bi} = m_b v_{bf} + M v_s \quad (\text{momentum})$$

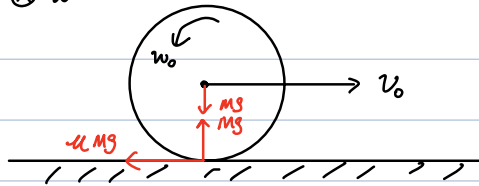
$$m_b v_{bi} r = m_b v_{bf} r + I w \quad (\text{angular momentum})$$

$$\frac{1}{2} m_b v_{bi}^2 = \frac{1}{2} m_b v_{bf}^2 + \frac{1}{2} M v_s^2 + \frac{1}{2} I w^2 \quad (\text{energy})$$

$\{ v_{bf}, v_s, w \}$ to be solved.

Example. Rotation with slipping.

⊗ $\hat{w}$



Q: time when slipping is finished?

v_{cm} when slipping is finished?

$$\mu Mg = Ma$$

$$\Rightarrow v_{cm} = v_0 - \mu g t \quad (1)$$

$$\mu Mg R = I \alpha$$

$$\Rightarrow \omega = -\omega_0 + \frac{\mu M g R}{I} t \quad (2)$$

$$v_{cm} = \omega R \quad (3)$$

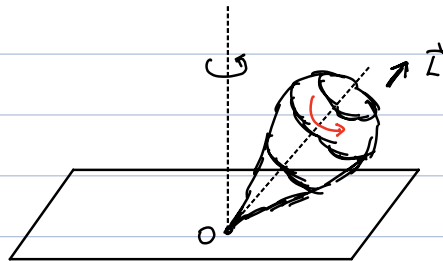
$$(1) + (2) + (3) \Rightarrow t = \frac{v_0 + \omega_0 R}{\mu g \left(1 + \frac{MR^2}{I}\right)}$$

$$I = \frac{1}{2} MR^2 \text{ for solid cylinder}$$

$$\Rightarrow t = \frac{v_0 + \omega_0 R}{3\mu g}$$

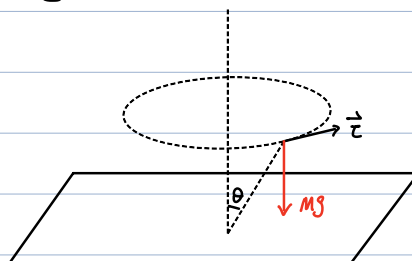
$$\Rightarrow v_{cm} = \frac{1}{3} (2v_0 - \omega_0 R)$$

Precession of gyroscope:



$$\vec{L} = \underbrace{\vec{r}_{cm} \times \vec{p}_{cm}}_{\approx 0 \text{ (fast spin)}} + \sum_i \vec{r}_i' \times \vec{p}_i'$$

trajectory of CM:



$$\vec{L} = \vec{r}_{cm} \times M\vec{g}$$

$$\Rightarrow M g r \sin \theta = \frac{dL}{dt}$$

$$dL = L \sin \theta d\phi$$

$$\Rightarrow M g r \sin \theta = L \sin \theta \frac{d\phi}{dt}$$

$$\Rightarrow \frac{d\phi}{dt} = \frac{M g r}{L}$$

(1)